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TrustField

Topology-Aware Ensemble Trust Propagation Intelligence

Verification & Containment System with Cyber-Physical Guards and 3D Holographic Visualization

Team No: PS-11

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Introduction

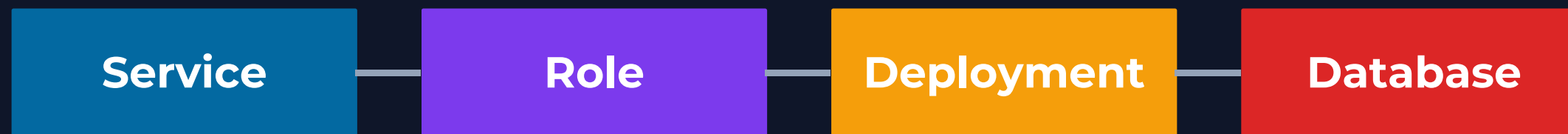


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The Challenge

Digital infrastructure relies on interconnected trust chains between services, roles, workloads, and secrets. An attacker can step through these chains — escalating privileges without ever having direct admin access.



Attack Escalation Path

TrustField Answers:

-  Maps entire infra as trust network
-  Simulates compromise spread
-  Verifies real exploitability
-  Installs cyber-physical guards
-  Visualizes in 3D holographic space

Problem Statement

To build a system that models trust relationships across infrastructure and predicts how attacks propagate through them, addressing the current inability of security tools to simulate cascading privilege compromise and reveal the true blast radius of a breach.



Literature Survey - Part 1 (Papers 1-5)



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#	Paper / Author	Domain	Key Contribution	Relevance to TrustField
1	Koch et al. (2005) J. Comp. & Sys. Sci.	Access Control	Graph-based specification of access control policies using directed graphs ($G=V,E$)	Foundational model for TrustField trust graph construction
2	Yin et al. (2024) World Wide Web	Graph ML	Heterogeneous graph semi-supervised learning for access control decisions	Heterogeneous node modeling: users, services, roles, workloads
3	Zhou et al. (2025) IEEE JSAC	Zero Trust	Decentralized federated graph learning with lightweight Zero Trust architecture for next-gen security	Zero Trust edge enforcement for delegation chain policy
4	Mohamed et al. (2021) SN Comp. Sci.	Authorization	Extended authorization policy for graph-structured data with complex relationship types	Edge-type modelling: AssumeRole, TokenMint, SecretRead, DeployTo
5	Rizvi & Fong (2020) ACM Trans. Priv. Sec.	Graph DB Auth	Efficient authorization of graph-database queries in attribute-supporting ReBAC model	Attribute-based graph query auth for workload and secret nodes



Literature Survey - Part 2 (Papers 6-10)



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#	Paper / Author	Domain	Key Contribution	Relevance to TrustField
6	Mahabadi et al. (2021) IEEE Access	Network Resilience	Network properties for robust multilayer infrastructure systems via percolation theory review	Percolation model for giant compromise cluster detection in trust graphs
7	Nwokoye & Madhusudanan (2022) Wireless Pers. Comm.	Epidemic Modeling	In-depth review of epidemic models for malicious-code propagation in wireless sensor networks	Epidemic model $I(t+1)=\beta AIt$ for trust cascade growth simulation
8	Mienye & Sun (2022) IEEE Access	Ensemble Learning	Survey of ensemble learning: concepts, algorithms, applications and prospects	Theoretical basis for weighted ensemble predictor: $\Sigma(w_i \cdot M_i)$
9	Wang et al. (2019) Cybersecurity	Exploit Analysis	From proof-of-concept to exploitable: real-world exploitability analysis and vulnerability pipeline	ExploitabilityGap = PBR - VBR framework design and verification logic
10	Shakeel et al. (2022) IEEE Access	Visualization	Comprehensive state-of-the-art survey on data visualization tools, challenges and domain-specific frameworks	Informs 3D trust field visualization platform selection rationale



Research Gaps Identified



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01

No Multi-Model Ensemble for Trust Propagation

Works use single models — epidemic OR percolation OR spectral — none combine all with topology-adaptive weighting.

02

Theoretical Blast Radius $\neq$ Real Exploitability

Attack graph tools compute theoretical blast radii but never verify which paths are actually exploitable. The Exploitability Gap (PBR-VBR) is unexplored.

03

Absence of Cyber-Physical Containment Hardware

Software-only containment is bypassable when the host OS is compromised. No system deploys TPM/FPGA guards at trust transition checkpoints.

04

Static Topology Assumptions in Propagation Models

Existing models assume homogeneous graphs. Topology fingerprinting and adaptive model selection for hub, chain, and dense-cluster topologies is absent.

05

No 3D Dynamic Trust Cascade Visualization

Security dashboards are 2D and static. Real-time 3D holographic visualization of trust propagation waves and containment barriers is entirely absent.



Objectives



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Trust Graph Modeling

Model entire infrastructure as a **directed graph ($G = V, E$)** with nodes for users, services, roles, workloads & secrets; edges for AssumeRole, TokenMint, SecretRead, DeployTo relationships.



Multi-Model Propagation

Simulate compromise spread using **Graph Traversal, Epidemic, Spectral Cascade, Percolation**, and **Control-System stability** models simultaneously.



Ensemble Prediction

Combine model predictions using a **weighted ensemble ($\sum w_i M_i$)** with adaptive weight adjustment over time based on accuracy.



Exploitability Verification

Perform controlled authentication traversal to compute **Exploitability Gap = PBR - VBR**, distinguishing theoretical risk from real-world exploitability.



Cyber-Physical Containment

Deploy **hardware guard modules** (STM32/TPM/FPGA) at high-risk trust transition points enforcing cryptographic token validation.



3D Holographic Visualization

Visualize trust propagation dynamics in **3D space** using holographic projection, LED volumes, or AR/VR, showing cascade waves and containment barriers.



Methodology — Six-Layer Pipeline



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1

Trust Graph Construction



$G=(V,E)$ with typed nodes & edges. Topology fingerprinting via clustering coefficient, centrality variance & spectral gap.

2

Multi-Model Propagation Engine



5 models simultaneously: BFS/DFS traversal, Epidemic (β_{Alt}), Spectral ($\tau > 1/\lambda_{max}$), Percolation (giant cluster), Control-System ($Ax+Bu$).

3

Ensemble Prediction & Topology



Voting + weighted ensemble $\Sigma(w_i M_i)$. Topology fingerprint selects best model: hub→spectral, chain→epidemic, dense→percolation.

4

Exploitability Verification



Controlled auth traversal generates tokens, simulates role assumptions. Produces PBR vs VBR. Exploitability Gap = PBR - VBR.

5

Cyber-Physical Containment (ECE)



STM32/TPM/FPGA guards validate: token authenticity, delegation depth, timestamp. 2-of-3 consensus. HW-SW feedback on risk elevation.

6

3D Holographic Visualization



Node size = privilege; brightness = cascade risk; spatial grouping = clusters. Propagation wave + containment barriers in real-time AR/VR.

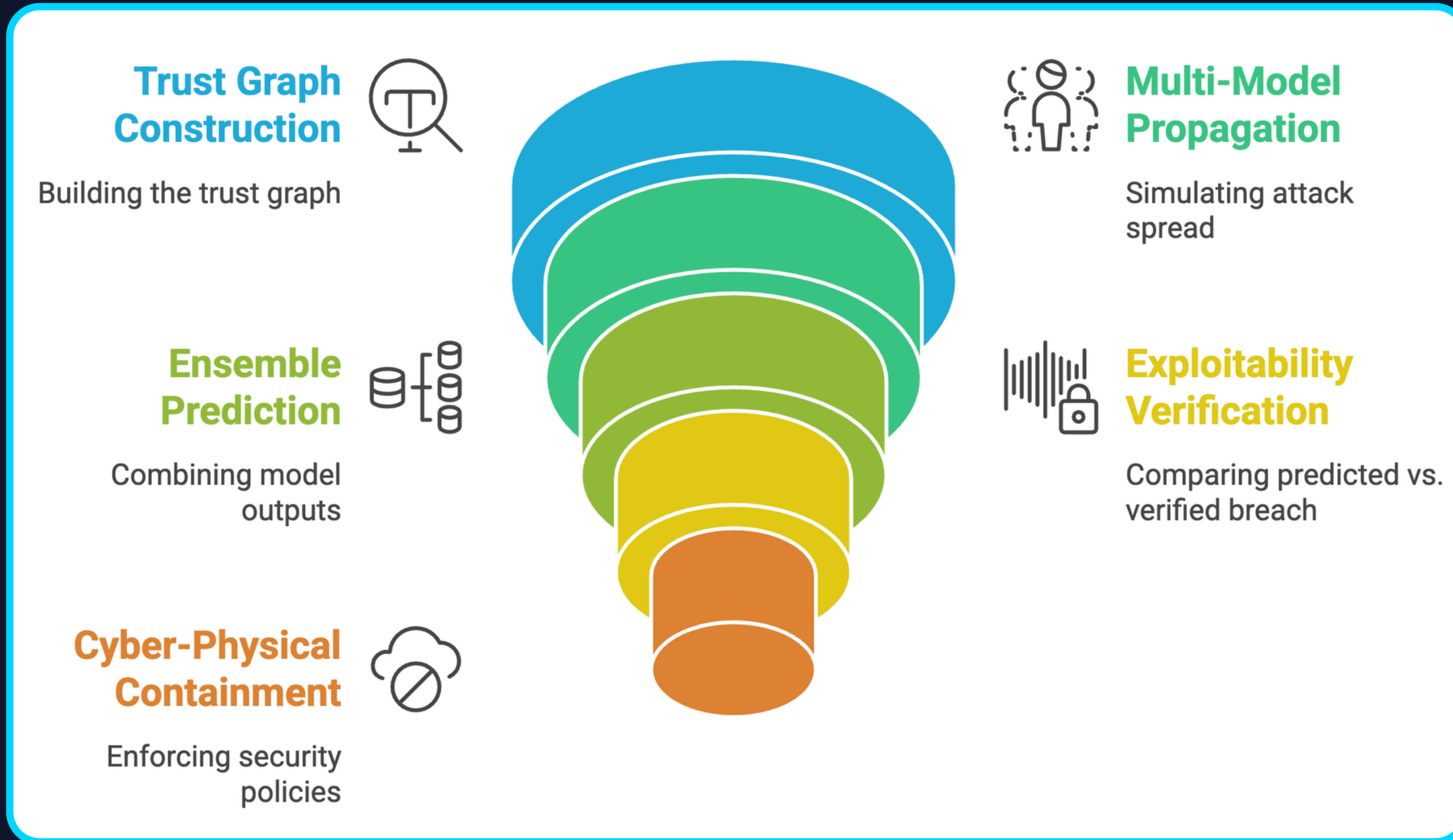


Methodology — Six-Layer Pipeline



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Software Requirements



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Graph Engine

- ▶ NetworkX (Python) or custom directed graph library
- ▶ Trust topology fingerprinting module
- ▶ Support for weighted directed edges and traversal

Propagation Models

- ▶ BFS/DFS reachability engine
- ▶ Spectral analysis (NumPy/SciPy)
- ▶ Percolation & control-system modules
- ▶ SIR epidemic simulator

Ensemble & ML

- ▶ scikit-learn or PyTorch for weighted aggregation
- ▶ Optional GNN for learned propagation patterns
- ▶ Model accuracy tracking pipeline
- ▶ Adaptive weight tuning module

Verification & IAM

- ▶ Delegation token generator
- ▶ Cloud IAM sandbox (AWS/GCP emulation)
- ▶ Exploitability Gap calculator (PBR-VBR)
- ▶ IAM role assumption simulator

Visualization

- ▶ 3D graph layout engine (Three.js / WebGL)
- ▶ Real-time propagation wave animation
- ▶ Containment barrier visualization layer
- ▶ Holographic or AR/VR rendering adapter

Infrastructure

- ▶ Docker/K8s test environment
- ▶ CI/CD pipeline simulation
- ▶ Microservice IAM test harness
- ▶ Logging & metrics collection (Prometheus)



Hardware Requirements



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Deployment Points



Identity Federation Gateways



Role Assumption Services



Secret Management APIs



Deployment Pipelines / CI-CD



Service Mesh Sidecar Proxies

Supported Hardware Platforms

STM32 Secure microcontroller

ESP32 With secure element

TPM 2.0 Trusted Platform Module

FPGA Policy engine (Xilinx/Intel)

Each Guard Module Contains



Secure key storage



Cryptographic validation engine



Local policy enforcement logic



Propagation monitoring sensors



Consensus: 2-of-3 guard approval



Optional FPGA acceleration



Interdisciplinary Relevance



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Computer Science

Graph Theory

Directed graph modeling, centrality, spectral analysis, percolation

Machine Learning

Ensemble methods, GNN for propagation pattern learning, adaptive weights

Distributed Systems

Cloud IAM, microservice security, multi-node trust federation

Algorithms

BFS/DFS traversal, epidemic SIR models, control-system analysis

Cybersecurity

Attack graph analysis, privilege escalation, blast radius verification



Electrical & Computer Eng.

Secure Hardware Design

STM32, ESP32, TPM 2.0 secure microcontrollers and elements

FPGA Acceleration

Near real-time graph traversal and policy enforcement at scale

Embedded Systems

Low-power guard modules with local cryptographic validation

Sensor Networks

Distributed behavioral monitoring, anomaly signal forwarding

Control Systems

Closed-loop hardware-software feedback for adaptive containment



Expected Outcomes



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>90%



Prediction Accuracy

Ensemble compromise prediction accuracy across hub, chain & dense cluster topology types

<15%



Exploitability Gap

Difference between Predicted Blast Radius (PBR) and Verified Blast Radius (VBR) post-calibration

>95%



Containment Rate

Successful blocking of high-risk trust transitions by cyber-physical hardware guards across all test environments

<100ms[⚡]

Enforcement Latency

Near real-time token validation latency via FPGA acceleration for large-scale distributed infrastructure

<5%



False Positive Rate

Minimized false alerts via topology-aware model selection and ensemble verification with adaptive weights

6



Novel Contributions

Cross-domain propagation, topology-aware ensemble, verified exploitability, cyber-physical containment, 3D viz, adversarial topology testing



Project Timeline — Gantt Chart (3 Months)





References



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- [1] Koch, Mancini, Parisi-Presicce. Graph-based specification of access control policies. J. Comp. & Sys. Sci. 71(1), 2005.
- [2] Yin et al. Heterogeneous graph semi-supervised learning for access control. World Wide Web 27, 35, 2024.
- [3] Zhou et al. Decentralized Federated Graph Learning with Zero Trust for Next-Gen Security. IEEE JSAC 43(6), 2025.
- [4] Mohamed et al. Extended Authorization Policy for Graph-Structured Data. SN Comp. Sci. 2, 351, 2021.
- [5] Rizvi & Fong. Efficient Authorization of Graph-DB Queries in ReBAC. ACM Trans. Priv. Sec. 23(4), 2020.
- [6] Mahabadi, Varga, Dolan. Network Properties for Robust Multilayer Infrastructure via Percolation. IEEE Access 9, 2021.
- [7] Moore & Cho. Applying Percolation Theory. Springer: Cyber Resilience of Systems & Networks, 2019.
- [8] Nwokoye & Madhusudanan. Epidemic Models of Malicious-Code in WSNs. Wireless Pers. Commun. 125, 2022.
- [9] Awasthi et al. Epidemic model for malware propagation in rechargeable WSN. J. Discrete Math. Sci. 24(5), 2021.
- [10] Zeng et al. Survey of attack graph analysis from data & knowledge perspective. Sec. & Comm. Networks, 2019.
- [11] Chatzimpampas et al. StackGenVis: Stacking Ensemble Learning. IEEE TVCG 27(2), 2021.
- [12] Ebrahimi & Zhang. Per-pixel accuracy weighting for ensemble classifiers. Int. J. Appl. Earth Obs. 122, 2023.
- [13] El-kenawy et al. Improved weighted ensemble for evapotranspiration prediction. Environ. Sci. Pollut. Res. 29, 2022.

- [14] Eskandari et al. Weighted ensemble for autonomous PV fault diagnosis. IJEPES 144, 2023.
- [15] Mienye & Sun. Survey of Ensemble Learning: Concepts & Applications. IEEE Access 10, 2022.
- [16] Wang et al. From proof-of-concept to exploitable. Cybersecurity 2, 12, 2019.
- [17] Jamshidi. AEGIS: Autonomous Exploit Generation & Intelligence System. Authorea Preprints, 2025.
- [18] Yuan et al. Exploitability Analysis via Taint Analysis. ICSP, 2022.
- [19] Yoon et al. Vulnerability Assessment Based on Real-World Exploitability. CSNet, 2023.
- [20] Lyu et al. Character-Level CNN for Predicting Exploitability. TASE, 2021.
- [21] Felicetti et al. RedToken for Hardware-Assisted Pre-Authentication. BCCA, 2025.
- [22] Wang et al. Predictive Containment Control for Multi-Agent Systems. IEEE TASE 22, 2025.
- [23] Shakeel et al. Comprehensive Survey on Data Visualization Tools. IEEE Access 10, 2022.
- [24] Korkut & Surer. Visualization in Virtual Reality: A Systematic Review. Virtual Reality 27, 2023.
- [25] Liu et al. Interactive Extended Reality in Information Visualization. IEEE THMS 52(6), 2022.